

Erik Alonso, Jesús Ruiz, Elisabete Aramendi, Digna González-Otero, Sofía Ruiz de Gauna, Unai Ayala, James K Russell, Mohamud Daya. Reliability and accuracy of the thoracic impedance signal for measuring cardiopulmonary resuscitation quality metrics. *Resuscitation*, Volume 88, 2015, Pages 28-34, ISSN 0300-9572, <https://doi.org/10.1016/j.resuscitation.2014.11.027>.

(<https://www.sciencedirect.com/science/article/pii/S0300957214008818>)

Abstract

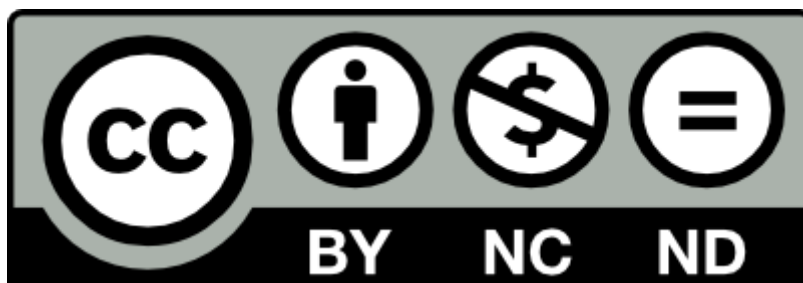
Aim: To determine the accuracy and reliability of the thoracic impedance (TI) signal to assess cardiopulmonary resuscitation (CPR) quality metrics.

Methods: A dataset of 63 out-of-hospital cardiac arrest episodes containing the compression depth (CD), capnography and TI signals was used. We developed a chest compression (CC) and ventilation detector based on the TI signal. TI shows fluctuations due to CCs and ventilations. A decision algorithm classified the local maxima as CCs or ventilations. Seven CPR quality metrics were computed: mean CC-rate, fraction of minutes with inadequate CC-rate, chest compression fraction, mean ventilation rate, fraction of minutes with hyperventilation, instantaneous CC-rate and instantaneous ventilation rate. The CD and capnography signals were accepted as the gold standard for CC and ventilation detection respectively. The accuracy of the detector was evaluated in terms of sensitivity and positive predictive value (PPV). Distributions for each metric computed from the TI and from the gold standard were calculated and tested for normality using one sample Kolmogorov-Smirnov test. For normal and not normal distributions, two sample t-test and Mann-Whitney U test respectively were applied to test for equal means and medians respectively. Bland-Altman plots were represented for each metric to analyze the level of agreement between values obtained from the TI and gold standard.

Results: The CC/ventilation detector had a median sensitivity/PPV of 97.2%/97.7% for CCs and 92.2%/81.0% for ventilations respectively. Distributions for all the metrics showed equal means or medians, and agreements >95% between metrics and gold standard was achieved for most of the episodes in the test set, except for the instantaneous ventilation rate.

Conclusion: With our data, the TI can be reliably used to measure all the CPR quality metrics proposed in this study, except for the instantaneous ventilation rate.

Keywords: Automated external defibrillator; Cardiopulmonary resuscitation quality; Chest compression; Thoracic impedance; Ventilations.



Reliability and accuracy of the thoracic impedance signal for measuring cardiopulmonary resuscitation quality metrics

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Word counts:

Abstract: 258 words

Paper : 3085 words

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Methods: A dataset of 63 out-of-hospital cardiac arrest episodes containing the compression depth (CD), capnography and TI signals was used. We developed a chest compression (CC) and ventilation detector based on the TI signal. TI shows fluctuations due to CCs and ventilations. A decision algorithm classified the local maxima as CCs or ventilations. Seven CPR quality metrics were computed: mean CC-rate, fraction of minutes with inadequate CC-rate, chest compression fraction, mean ventilation rate, fraction of minutes with hyperventilation, instantaneous CC-rate and instantaneous ventilation rate. The CD and capnography signals were accepted as the gold standard for CC and ventilation detection respectively. The accuracy of the detector was evaluated in terms of sensitivity and positive predictive value (PPV). The reliability of the TI to measure CPR quality metrics was analyzed by comparing statistically the metrics obtained from the TI with those obtained from the gold standard. The accuracy of each metric was analyzed individually for each episode.

Results: The CC/ventilation detector had a median sensitivity/PPV of 97.2%/97.7% for CCs and 92.2%/81.0% for ventilations respectively. Distributions for all the metrics showed equal means or medians, and agreements $> 95\%$ between metrics and gold standard was achieved for most of the episodes in the test set, except for the instantaneous ventilation rate.

Conclusion: With our data, the TI can be reliably used to measure all the CPR quality metrics proposed in this study, except for the instantaneous ventilation rate.

Keywords

Cardiopulmonary Resuscitation Quality, Thoracic Impedance, Chest Compression, Ventilations, Automated External Defibrillator

1. INTRODUCTION

Delivery of high quality cardiopulmonary resuscitation (CPR) is a key component of current resuscitation guidelines.^{1,2} Minimally interrupted chest compressions (CCs), with a rate of at least 100 min⁻¹ and at least 5 cm depth, are recommended along with allowance for full chest recoil on completion of each compression. Guidelines also recommend oxygenation to be provided with a ventilation rate of 8-10 min⁻¹ in intubated patients. Several studies have revealed that low quality CCs, pauses in CCs, and hyperventilation are associated with poorer outcomes in both animals³⁻⁹ and humans.¹⁰⁻¹⁴ Nevertheless, suboptimal quality of CPR is common in both in-hospital and out-of-hospital cardiac arrest (OHCA).^{7,9,12-18}

Procedures such as episode debriefing or the incorporation of feedback systems into defibrillators have been developed to improve CPR quality. A standardized review of resuscitation episodes permits reporting of rescuer performance in terms of CPR quality metrics related to CCs and ventilations. Efforts have been made to streamline these reports by defining the CPR quality metrics that should be reported.¹⁹ Systems for real time feedback on CPR have proven effective to improve metrics.²⁰ The feedback on too slow CCs or hyperventilation can help guide rescuers towards metrics recommended by guidelines.^{16,21-24}

In the context of advanced life support (ALS), the monitor/defibrillators may be equipped with acceleration/force sensors as well as capnography and pulse oximetry modules. These may be used to assist the rescuer on the CPR quality. By contrast, in the context of basic life support (BLS), some automated external defibrillators (AEDs) also offer acceleration/force sensors for feedback, but these are relatively expensive accessories for widespread use. Often the only patient interface is defibrillation pads, and only the electrocardiogram (ECG) and TI signals are available.

Every CC causes a fluctuation visible in the TI signal. The TI signal can be used to identify CCs^{16,25,26} and ventilations.^{16,27-30} Algorithms for CC detection using the TI signal are integrated into commercial software for episode reviewing, and algorithms for ventilation detection have been recently proposed using the TI.^{27,28} Their global performance is provided in terms of sensitivity and positive predictive value (PPV).²⁵ Nevertheless, detailed analysis of the utility of the TI signal for CC/ventilation metrics is needed.

The purpose of this study was to analyze retrospectively OHCA episodes to determine the reliability and accuracy of using the TI to evaluate seven quality metrics of the CCs and the

31 ventilations. An efficient detector for CCs and ventilations was developed, and the reliability and
32 accuracy of each of the seven CPR metrics was evaluated globally and for each patient.

2. MATERIALS AND METHODS

2.1. Data materials

The dataset used in this study was a subset of a large out-of-hospital cardiac arrest (OHCA) registry containing 623 episodes maintained by the Tualatin Valley Fire & Rescue agency (Tigard, Oregon, USA). The episodes, one per patient, were collected using the Philips HeartStart MRx monitor/defibrillator between 2006 and 2009. From the original database only 199 episodes had the compression depth signal (CD), the TI signal and the capnogram. Those episodes where the three signals were concurrent for at least 20 min were selected. A total of 63 episodes with mean (SD) duration 41 (11) min comprised the dataset. The CD signal (sampling rate 250 Hz) was computed from the force and the acceleration recorded through a CPR assist pad. The TI signal was recorded through the defibrillation pads by applying a sinusoidal excitation current (32 kHz, 3 mA peak-to-peak) with a resolution of 0.74 mΩ per least significant bit, a bandwidth of 0 – 80 Hz and a sampling rate of 200 Hz. The capnogram was acquired using Microstream (sidestream acquisition) with a frequency rate of 40 Hz and a resolution of 0.004 mmHg per bit.

For the CC-detection the CD was used as gold standard. The instants of CCs were automatically marked applying a fixed threshold of -15 mm and manually reviewed (panel a in Fig. 1). For ventilation detection, the capnogram was used as gold standard and ventilations were annotated independently by three experienced biomedical engineers relying on visual inspection (panel a in Fig. 1). Intervals where capnography was disconnected (0.90% of the time) or uninterpretable because the ventilation pattern was not clearly recognizable (11.65% of the time), were excluded from the analysis. A total of 2575 minutes were analyzed, which included 110286 CCs and 17586 ventilations. Episodes were randomly allocated to training and test sets, 32 and 31 episodes respectively.

2.2. Simultaneous CC/ventilation detector

An algorithm that simultaneously detects CCs and ventilations was developed based on the TI signal. The TI signal, $z[n]$, where n is the time sample number, was digitally preprocessed. Local maxima were identified, and waveform features characterizing the signal in their vicinity were extracted. A decision algorithm based on the waveform features determined whether or not each local maximum was CC or ventilation (Fig. 2).

Signal preprocessing

The $z[n]$ signal was preprocessed as shown in Fig. 2. For CC detection, a low-pass filter with a cut-off frequency of $f_{c_1} = 1.8\text{Hz}$ was applied in order to remove high frequency noise and to retain fluctuations due to CCs and ventilations. The resulting signal is denoted as $z_c[n]$. For ventilation detection, $z[n]$ was low-pass filtered at a cut-off frequency of $f_{c_2} = 0.6\text{Hz}$ aiming to suppress fluctuations due to CCs as well as high frequency noise. In this case, the signal obtained is expressed as $z_v[n]$.

Peak detection and feature extraction

The $z_c[n]$ and $z_v[n]$ signals were analyzed independently to detect the instants of the local maxima. For each local maximum detected in $z_c[n]$ and $z_v[n]$, which might be a potential CC or ventilation respectively, the following waveform features were computed:

- A_1 : The trough-to-peak amplitude of the fluctuation.
- A_2 : The peak-to-trough amplitude of the fluctuation.
- d_1 : The duration of the trough-to-peak rise.
- d_2 : The duration of the peak-to-trough fall.

The features and locations of each fluctuation in $z_c[n]$ and $z_v[n]$ were stored in the vectors v_c and v_v respectively (Fig. 2). Panel b in Fig. 1 illustrates, from top to bottom respectively, examples of $z[n]$, $z_c[n]$ and $z_v[n]$ signals, where the extracted features are depicted.

Decision algorithm

For CC detection, the decision algorithm decided whether a local maximum was considered as CC, based on the extracted features and location of each fluctuation, v_c . The algorithm evaluated the duration of the fluctuation ($d_1 + d_2$) against a static threshold and the mean amplitude, mean value of A_1 and A_2 , against a dynamic threshold given by:

$$Th_c = W_c \cdot \sum_{k=1}^{N_c} \frac{A_{1k} + A_{2k}}{2} \quad (1)$$

Th_c represents the weighted average of the mean amplitude of the last 6 detected CCs, and W_c denotes the weighting factor. The fluctuation was classified as CC if its duration and mean

amplitude were above the thresholds established by the decision algorithm. A refractory period between consecutive CCs was considered to avoid possible false positives.

For ventilation detection, the decision algorithm classified a local maximum as ventilation based on the extracted features and location of each fluctuation, v_v . The decision algorithm evaluated the inflation time (d_1) against a static threshold, and the inflation amplitude, A_1 , against a dynamic threshold defined as follows:

$$Th_v = W_v \cdot \sum_{k=1}^{N_v} \min(A_{1k}, A_{2k}) \quad (2)$$

Th_v describes the weighted average of the minimum amplitude of the last 17 detected ventilations, and W_v denotes the weighting factor. The fluctuation was classified as ventilation if its inflation time and inflation amplitude were above the thresholds established by the decision algorithm. A refractory period between consecutive ventilations was considered in order to avoid possible false positives.

The instants of CCs and ventilations detected by the decision algorithm were denoted as t_c and t_v respectively (Fig. 2).

2.3. CPR quality metrics

Seven different CPR quality metrics were computed using t_c and t_v instants:

- **Mean CC-rate:** Mean frequency of CCs during CC-series. Consecutive CCs separated by less than 1.5 s formed a CC-series.¹⁹ A single value per episode was reported.
- **Fraction of minutes with inadequate CC-rate (FMIR):** Proportion of minutes with CC-rate below 90 cpm or above 120 cpm.¹⁹ A tolerance of 10 cpm from the lower limit (100 cpm) recommended by the 2010 resuscitation guidelines¹ was established to consider a CC-rate as inadequate. A single value per episode was reported.
- **Chest compression fraction (CCF):** Proportion of time without spontaneous circulation during which CCs were provided.³¹ A single value per episode was reported.
- **Mean ventilation rate:** Mean value of the ventilations provided every minute of the episode.¹⁹ A single value per episode was reported.

- **Fraction of minutes with hyperventilation (FMH):** Proportion of minutes with more than 15 ventilations.¹⁹ A single value per episode was reported.
- **Instantaneous CC-rate:** Rate computed every 1.5s as the median rate of the last 8 consecutive CCs.
- **Instantaneous ventilation rate:** Rate computed every 15s as the number of ventilations provided in the last minute.

2.4. Evaluation of the detector

The training set was used to develop the simultaneous CC/ventilation detector; performance was assessed with the test set in terms of sensitivity (proportion of correctly detected CC/ventilations) and PPV (proportion of detected CCs/ventilations corresponding to real CCs/ventilations).

2.5. Evaluation of the feasibility and accuracy of the TI signal to measure CPR quality metrics

The feasibility of the TI signal as a surrogate of the CD and capnogram signals to measure CPR quality metrics was evaluated. The CD signal was considered as gold standard to evaluate CC metrics and the capnogram to evaluate those metrics linked to ventilations.

Distributions for each metric obtained from the TI and from the gold standard were analyzed independently applying the one sample Kolmogorov-Smirnov test to assess normality. For normal distributions, Barlett's test and two sample t-test were performed to test for equal variances and equal means respectively. For not normal distributions, the Mann-Whitney U test was used to test for equal medians. A $p - \text{value} < 0.05$ was considered significant.

For each CPR quality metric, a Bland-Altman plot was represented to analyze the level of agreement (LOA) between the values obtained from the TI signal and those obtained from the gold standards.

For the instantaneous CC-rate, the percentage of rates with an error above 10% was computed for each episode, as well as the the percentage of false rates, i.e. percentage of rates that were reported inadequate (below 90 and above 120 cpm) by the TI signal but adequate (between 90 and 120 cpm) by the gold standard plus rates that were reported adequate by the TI signal but inadequate by the gold standard.

140 For the instantaneous ventilation rate, the percentage of rates with an error > 2 ventilations
141 per min was computed for each episode, as well as the percentage of false rates, i.e. percentage of
142 rates that were reported inadequate ($> 15 \text{ min}^{-1}$) by the TI signal but adequate ($\leq 15 \text{ min}^{-1}$) by
143 the gold standard, plus rates that were reported adequate by the TI signal but inadequate by the
144 gold standard.

3. RESULTS

Fig. 3 shows the performance of the simultaneous CC/ventilation detector. The box plots representing the sensitivity and PPV for CC and ventilation detection can be observed in panels a and b, respectively. The median (IQR) sensitivities were 97.2 (95.6-98.6) and 92.2 (87.4-95.8); and the median (IQR) PPV were 97.7 (94.8-98.5) and 81.0 (67.2-90.5) for CC and ventilation detection respectively.

Table 1 summarizes the mean (SD) values obtained from the gold standard and from the TI signal for each CPR quality metric as well as the mean (SD) of the unsigned error. The p – values obtained from either t-test or Mann-Whitney U test are also reported. Distributions for instantaneous CC-rate and FMH obtained from the TI and from the gold standard were not normal, but had equal medians ($p = 0.99$ and $p = 0.66$ respectively). The instantaneous ventilation rate did come from a normal distribution but did not show equal means ($p < 0.001$). The rest of CPR quality metrics came from normal distributions, where equal variances and equal means were reported by Barlett’s test and t-test respectively. The mean ventilation rate was assumed to come from normal distribution, although the p – value (0.07) was close to the significant level (0.05).

The Bland-Altman plots representing the agreement between the values obtained from the TI and from the gold standard for the mean CC-rate, CCF, FMIR, mean ventilation rate and FMH can be observed in Fig. 4.

Fig. 5 shows the Bland-Altman plots for the instantaneous CC-rate (panel a) and instantaneous ventilation rate (panel d). In any case most of the reported values (mean of 99.5% CC-rates per episode and mean of 97.7% ventilation rates per episode) are in the 95% LOA. For each of the 31 episodes the percentage of instantaneous CC-rates with errors $> 10\%$ is plotted in panel b, and the false rate in panel c. The mean error per episode was 1.8% and the mean false rate 1.2%. The percentage of instantaneous ventilation rates with errors $> 2 \text{ min}^{-1}$ is plotted for each episode in panel e, and the false rate in panel f. The mean error per episode was 26.0% and the mean false rate 6.8%.

4. DISCUSSION

A unified method to detect CCs and ventilations in the TI was developed. Reliability of using the TI to report on CPR metrics was examined in real life settings where multiple rescuers and patients were involved.

4.1. CC/ventilation detector: The gold standard

The detector was developed using entire cardiac arrest episodes including CCs and ventilations. For the CC-detection the CD was used as gold standard.

For ventilation detection Risdal et al.²⁷ considered as gold standard the fluctuations manually marked in TI, while Edelson et al.²⁸ used both, TI fluctuations and capnogram variations. In this study the capnogram was considered as gold standard. For patients in cardiac arrest, artifacts in the capnogram due to chest compressions and other factors, make the identification of ventilations difficult in some intervals.³² In our database 11.65% of the capnogram was uninterpretable, as in the intervals shown in Fig. 6 (supp.). These intervals were discarded for evaluation purposes.

As a limitation, the volume of the ventilations was not considered in this approach. Although the guidelines recommend ventilations of 7-10ml/kg of tidal volume, only the use of a flow monitoring system would provide such information.^{27,30}

4.2. CC/ventilation detector: The method

Our method unifies the detection of both CCs and ventilations in a single temporal method based on different preprocessing and adaptive thresholding, but common feature extraction. The overall results of our method, sensitivity/PPV of 97.2%/97.7% for CCs and 92.2%/81.0% for ventilations, are close to scores reported by other authors. Ayala et al.³³ reported sensitivity/PPV of 94.2%/97.4% with 38 cardiac arrest episodes, part of an inter-centre CPR quality study. Commercial software reported global sensitivity/PPV of 94%/90%.²⁵

For ventilation detection, Risdal et al.²⁷ obtained 94.8/88.7% and 95.5/90.4% in episodes with and without CCs respectively. They applied an adaptive algorithm to suppress CC-artifacts prior to applying neural networks and detecting fluctuations that had been manually marked in the TI. Similarly, Edelson et al.²⁸ applied an adaptive filtering to suppress artifacts and reported 78%/87% detecting ventilations in the TI. Our method provides similar results with a simpler processing.

4.3. Reliability of CPR quality metrics

The first five metrics analyzed in this study are computed one per episode. and can be used to review individual cardiac arrest episodes or for inter-centre CPR quality studies.

The global results (Table 1) confirm that the five CPR metrics computed from the TI are reliable for clinical trials: they showed equal means or medians when compared with the metrics obtained from the gold standard, and errors associated with any of them were minor.

An effective approach to improve resuscitation quality is the debriefing of each episode (patient) on the basis of a list of CPR quality metrics. The accuracy of each metric computed from the TI was analyzed for each patient. Fig. 4 showed a 95% LOA between the metric and the gold standard for most of the 31 episodes. One case was out of the 95% LOA for the mean CC-rate, three for the FMIR, two for the CCF and two for both ventilation metrics.

4.4. Accuracy of the instantaneous rates

The instantaneous rates for CCs and ventilations computed from the TI may be applied to give rate feedback when using AEDs which only acquire ECG and TI signals. Moreover, the time evolution of instantaneous metrics may be plotted for CPR quality analysis in debriefing 'report sheets'.

Table 1 showed that the instantaneous CC-rate computed from TI and from the gold standard came from not normal distributions with equal medians ($p = 0.99$); while the instantaneous ventilation rate showed normal distributions with different mean ($p < 0.001$). The Bland-Altman plots are shown in panels a and d of Fig. 5. Values of instantaneous CC-rate showed 95% LOA for most of the points, excluding those associated to episodes 13 and 22. In episode 13, illustrated in Fig. 7 (supp.), we observe that the fluctuations caused in the TI differ substantially in two different time intervals of the episode (panels a and b). This variability could be explained by rescuer alternation or changes in skin-electrode contact, and should be considered when used for episode debriefing. In global evaluation of the instantaneous ventilation rate, the values showed great dispersion in the range of LOA (between -8 and 5 ventilations) and several values out those limits.

In above mentioned applications, performance of each metric for every patient should be considered if debriefing based on the instantaneous rate is to be integrated. Fig. 5 showed the percentage of errors and false rates for each episode. Panel b showed that the percentage of error

229 $> 10\%$ was below 6% in all cases but one (episode 13) for rate of CCs. In contrast, for ventilations,
230 panel e showed that the mean of values with errors > 2 vent/min was 26%, with episodes where
231 the percentage was as high as 68%. Fig. 8 (supp.) illustrates intervals of episode 31 where the TI
232 showed fluctuations that the algorithm identified while the capnogram did not show any ventilation,
233 and they were wrongly classified. Attending to the percentage of false rates, the result for CCs was
234 below 7% in any episode (panel c in Fig. 5), while the percentage of false rates was higher than
235 20% in 7 out of 31 for ventilation (panel f in Fig. 5).

236 The inter-patient analysis showed that the TI can be reliable for instantaneous CC rate in most
237 of the episodes, while is not reliable for instantaneous ventilation rate in a considerable percentage.
238 These negative results may be aligned with those given by Edelson et al.²⁸, that reported 40% of
239 the rates with a deviation > 2 vent/min. With our CC/ventilation detector the TI is reliable for
240 mean ventilation rate because the negative effect of false positives is smoothed when a large amount
241 of ventilations are considered to compute the metric. That is not the case for the instantaneous
242 ventilation rate.

243 4.5. Limitations of the study

244 Results and conclusions of this study are closely linked to the specific features of the TI acquired
245 by the Philips HeartStart MRx monitor/defibrillator. Considering the TI from other equipment,
246 acquired with different electronic circuitry and defibrillation pads, might draw other conclusions.

247 5. Conclusions

248 In this study, a simultaneous CC/ventilation detector based on the TI signal has been developed
249 and the reliability and accuracy of the TI to measure CPR quality metrics analyzed. Metrics related
250 to CCs can be reliably used for reviewing individual cardiac arrest episodes, for inter-centre CPR
251 quality studies and for giving rate feedback when using AEDs. Although the ventilation metrics
252 are accurate, the instantaneous ventilation rate has unacceptable large errors in several episodes.

253 Conflict of interest

254 Authors, J. Ruiz, E. Aramendi, and S. Ruiz de Gauna have received research support from
255 Bexen Cardio (Ermua, Spain) for studies on AED shock advice algorithms. James K. Russell is
256 employed by Philips. Mohamud Daya is an unpaid consultant for Philips Healthcare.

257 **Acknowledgements**

258 This work received financial support from the Ministerio de Economía y Competitividad of
259 Spain through the projects TEC2012-31928 and TEC2012-31144, from the University of the Basque
260 Country (UPV/EHU) through the unit UFI11/16 and from the Programa de Formación de Personal
261 Investigador del Departamento de Educación, Universidades e Investigación del Gobierno Vasco
262 through the grants BFI-2010-174, BFI-2010-235 and BFI-2011-166.

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Figure Legends

- Figure 1 Examples of the signals comprised in each episode: CD, capnogram and TI signals from top to bottom in panel a. Reviewers' annotations of CCs and ventilations are depicted as red dotted lines in CD and capnogram signals respectively. Panel b shows (from top to bottom) in more detail the segment colored in blue in the TI signal of panel a, $z[n]$, the same segment processed to enhance CCs, $z_c[n]$, and the same segment processed to enhance ventilations, $z_v[n]$. The extracted features are also represented in both $z_c[n]$ and $z_v[n]$.
- Figure 2 Scheme used to compute the instants of both CCs and ventilations using exclusively the TI signal, $z[n]$.
- Figure 3 Box plots showing the performance of the simultaneous CC/ventilation detector. The sensitivity and PPV values for CCs and ventilations for the test set are depicted in panels a and b respectively.
- Figure 4 Bland-Altman plots for CC-rate, FMIR, CCF, ventilation rate and FMH are represented in a, b, c, d and e panels respectively. The 95% LOA is depicted in black dashed lines.
- Figure 5 Results obtained from the instantaneous rate analysis. a and d panels show the Bland-Altman plots for instantaneous CC-rate and instantaneous ventilation rate respectively. The percentage of instantaneous rates with an error above 10% in CC-rate and percentage of instantaneous rates with an error greater then 2 min^{-1} in ventilation rate are represented per episode in panels b and e. The percentage of false rates for instantaneous CC-rate and instantaneous ventilation rate are shown in panels c and f respectively.
- Figure 6 (Supp.) Examples of uninterpretable intervals in the capnogram. From top to bottom intervals with: questionable inflation/deflation pattern, CPR artifact, unusual waveform, and CPR artifact in the 10 – 40 s interval.
- Figure 7 (Supp.) Two different time intervals corresponding to the same episode are shown in panels a and b. The CD, $z[n]$, and $z_c[n]$ signals can be observed in both panels from top to bottom respectively. Red dotted lines depict the instants of the CCs marked in the gold standard. Red triangles represent the instants of the correctly detected CCs (true positives), whereas black triangles represent the instants of the incorrectly detected CCs (false positives). In $z_c[n]$ of panel a, the errors in the detection due to the second harmonic of the TI fluctuation are illustrated. However, there are no detection errors in panel b as there is no second harmonic in the TI.

Figure 8

(Supp.) Two different time intervals corresponding to the same episode are shown in panels a and b. The capnography, $z[n]$, and $z_v[n]$ signals can be observed in both panels from top to bottom respectively. Red dotted lines depict the instants of the ventilations marked in the gold standard. Red triangles represent the instants of the correctly detected ventilations (true positives), whereas black triangles represent the instants of the incorrectly detected ventilations (false positives). In $z_v[n]$ of panel a, there are no detection errors. However, there are two detection errors in panel b due to fluctuations in the TI not corresponding to ventilations as can be seen in the capnogram.

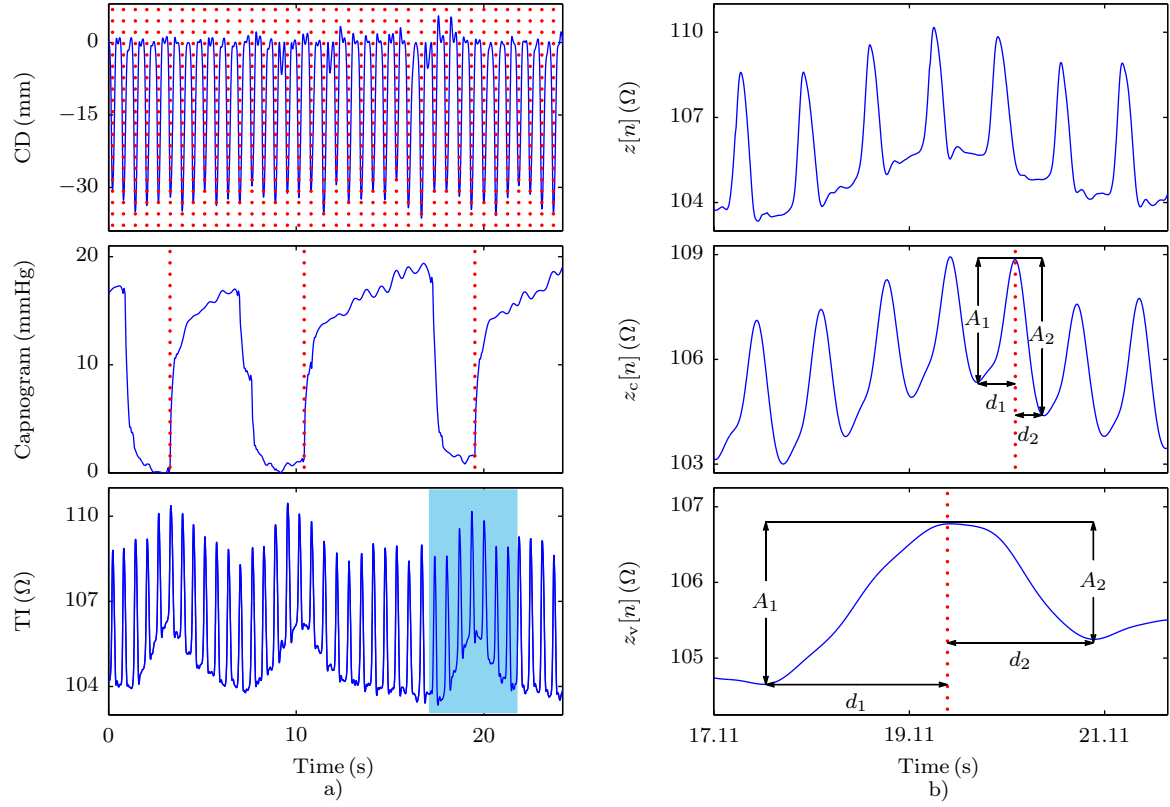


Figure 1: Examples of the signals comprised in each episode: CD, capnogram and TI signals from top to bottom in panel a. Reviewers' annotations of CCs and ventilations are depicted as red dotted lines in CD and capnogram signals respectively. Panel b shows (from top to bottom) in more detail the segment colored in blue in the TI signal of panel a, $z[n]$, the same segment processed to enhance CCs, $z_c[n]$, and the same segment processed to enhance ventilations, $z_v[n]$. The extracted features are also represented in both $z_c[n]$ and $z_v[n]$.

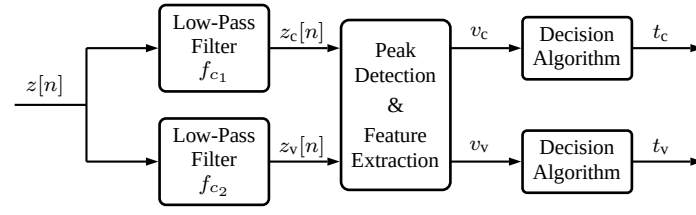


Figure 2: Scheme used to compute the instants of both CCs and ventilations using exclusively the TI signal, $z[n]$.

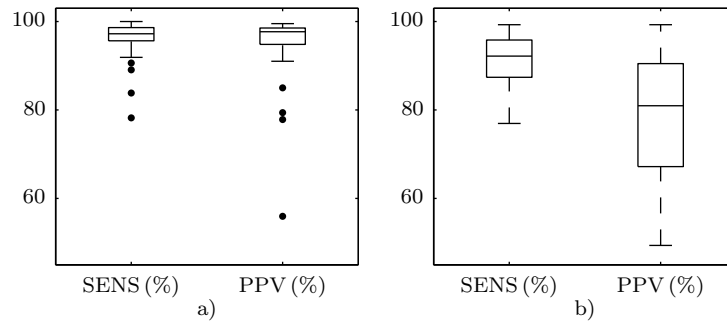


Figure 3: Box plots showing the performance of the simultaneous CC/ventilation detector. The sensitivity and PPV values for CCs and ventilations for the test set are depicted in panels a and b respectively.

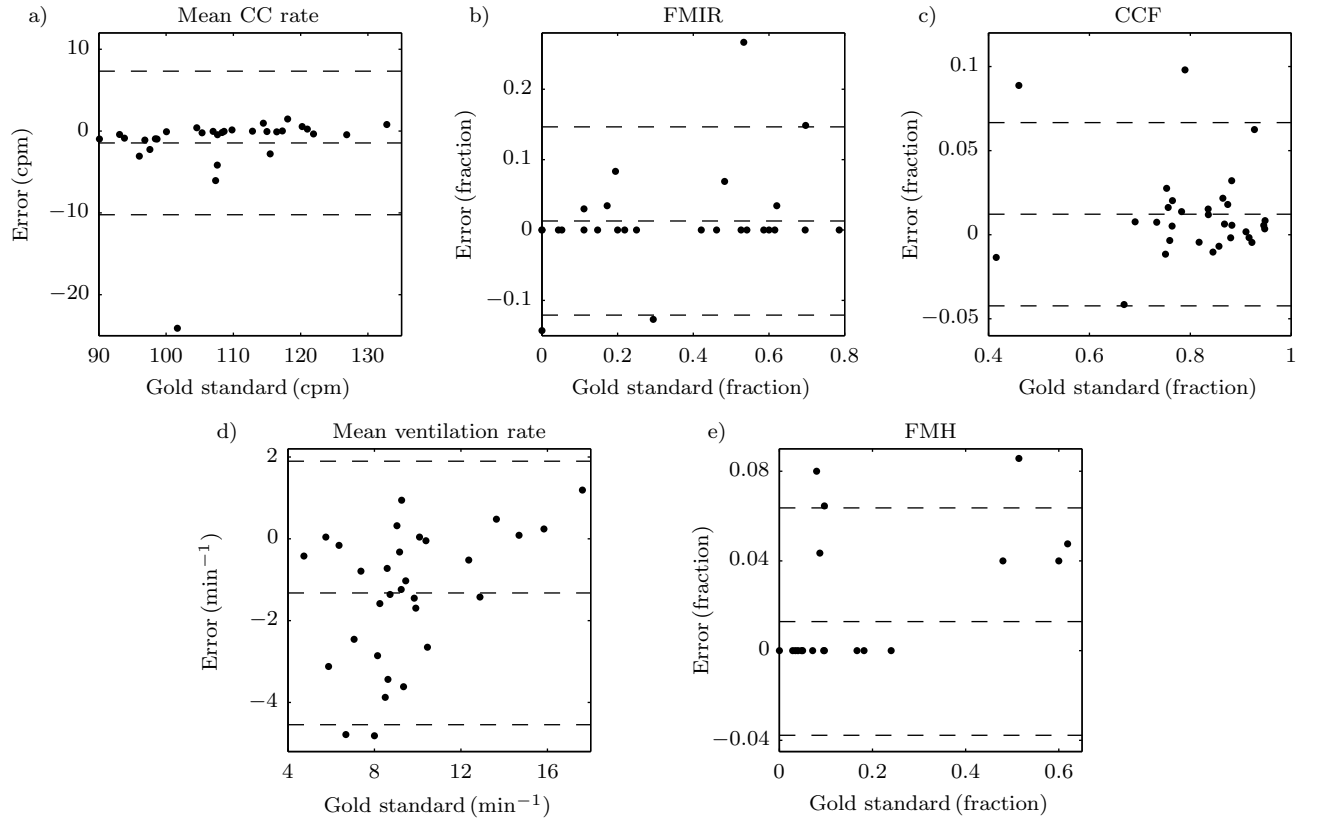


Figure 4: Bland-Altman plots for CC-rate, FMIR, CCF, ventilation rate and FMH are represented in a, b, c, d and e panels respectively. The 95% LOA is depicted in black dashed lines.

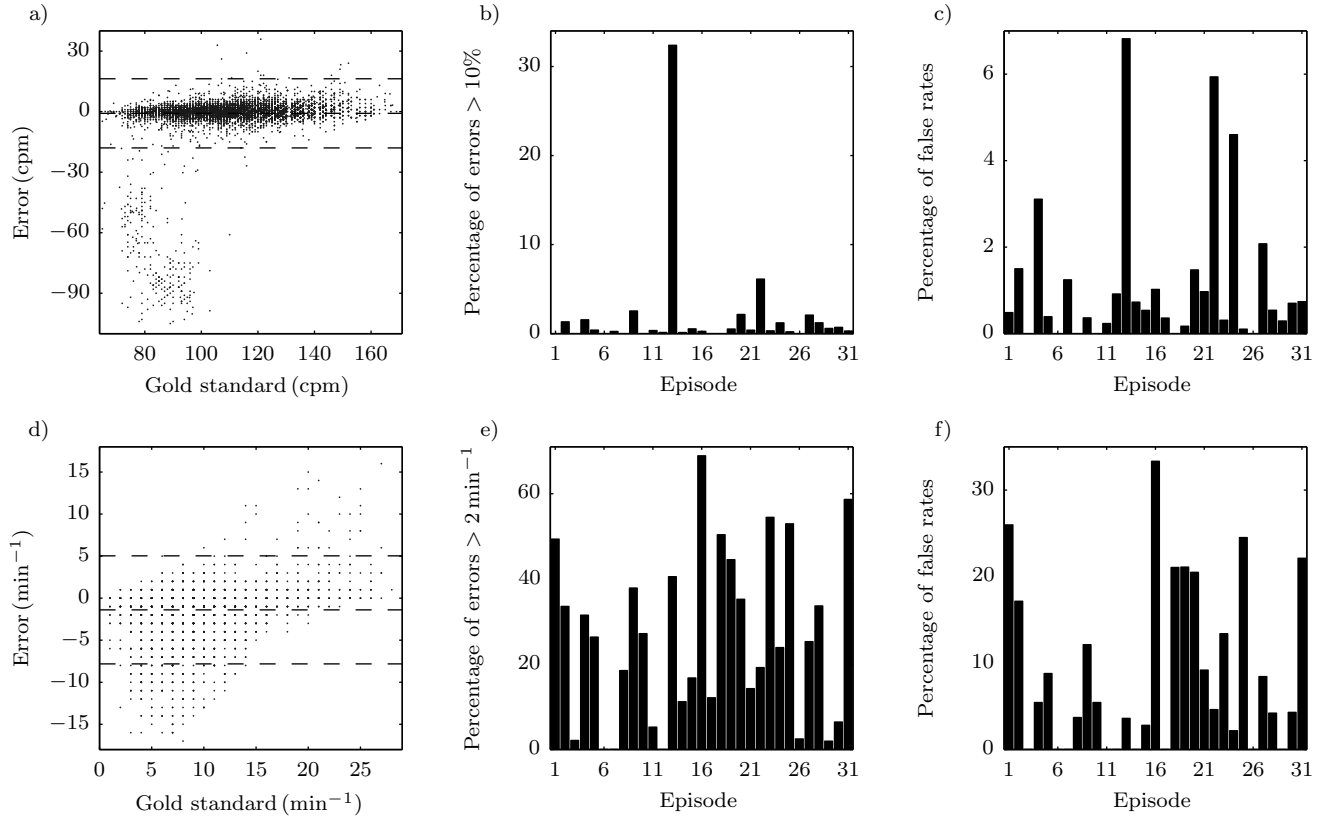


Figure 5: Results obtained from the instantaneous rate analysis. a and d panels show the Bland-Altman plots for instantaneous CC-rate and instantaneous ventilation rate respectively. The percentage of instantaneous rates with an error above 10% in CC-rate and percentage of instantaneous rates with an error greater then 2 min⁻¹ in ventilation rate are represented per episode in panels b and e. The percentage of false rates for instantaneous CC-rate and instantaneous ventilation rate are shown in panels c and f respectively.

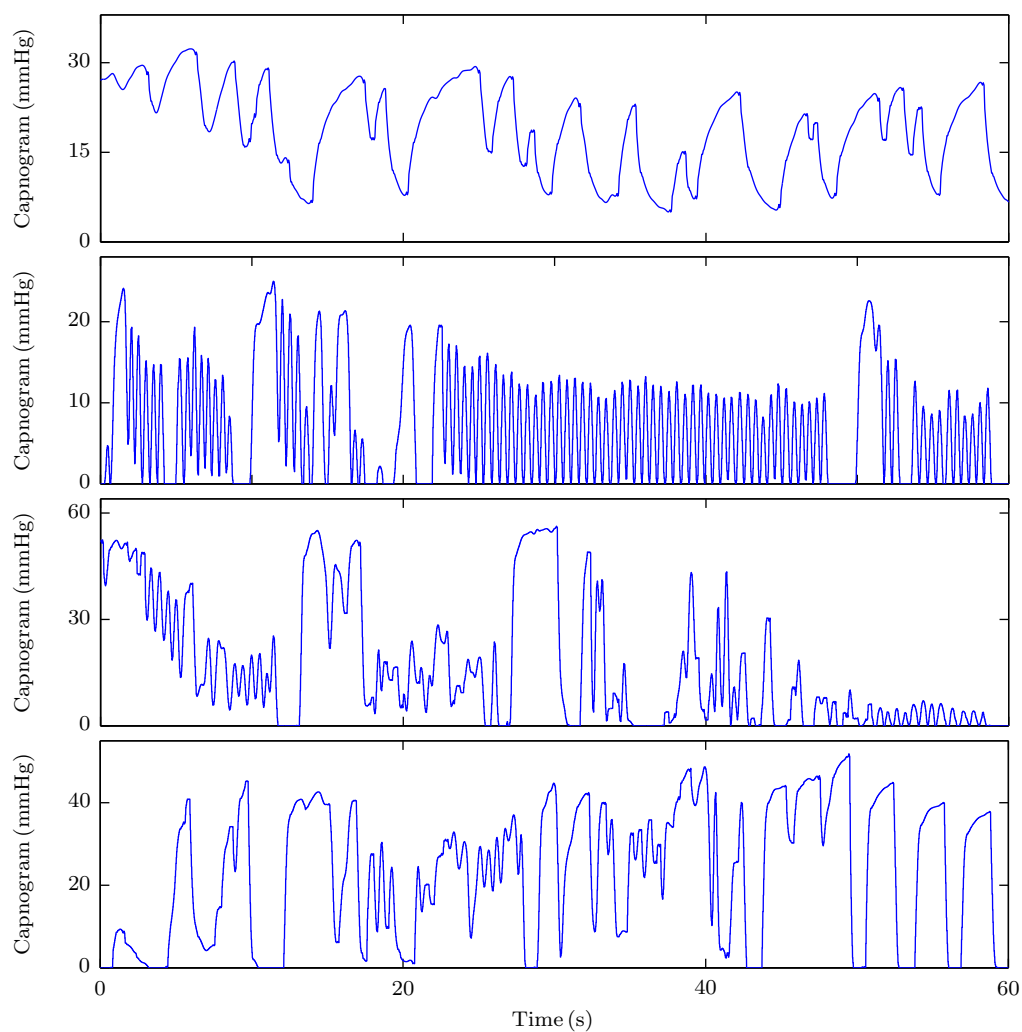


Figure 6: (Supp.) Examples of uninterpretable intervals in the capnogram. From top to bottom intervals with: questionable inflation/deflation pattern, CPR artifact, unusual waveform, and CPR artifact in the 10 – 40 s interval.

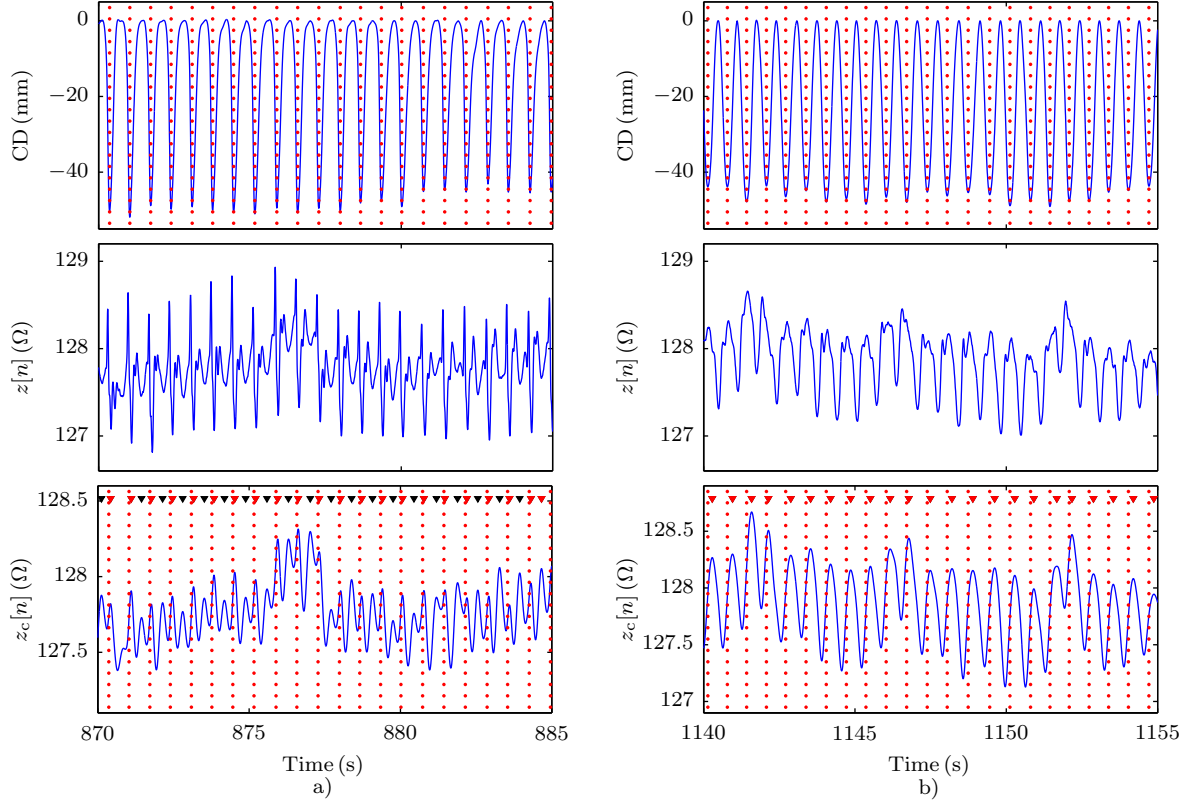


Figure 7: (Supp.) Two different time intervals corresponding to the same episode are shown in panels a and b. The CD, $z[n]$, and $z_c[n]$ signals can be observed in both panels from top to bottom respectively. Red dotted lines depict the instants of the CCs marked in the gold standard. Red triangles represent the instants of the correctly detected CCs (true positives), whereas black triangles represent the instants of the incorrectly detected CCs (false positives). In $z_c[n]$ of panel a, the errors in the detection due to the second harmonic of the TI fluctuation are illustrated. However, there are no detection errors in panel b as there is no second harmonic in the TI.

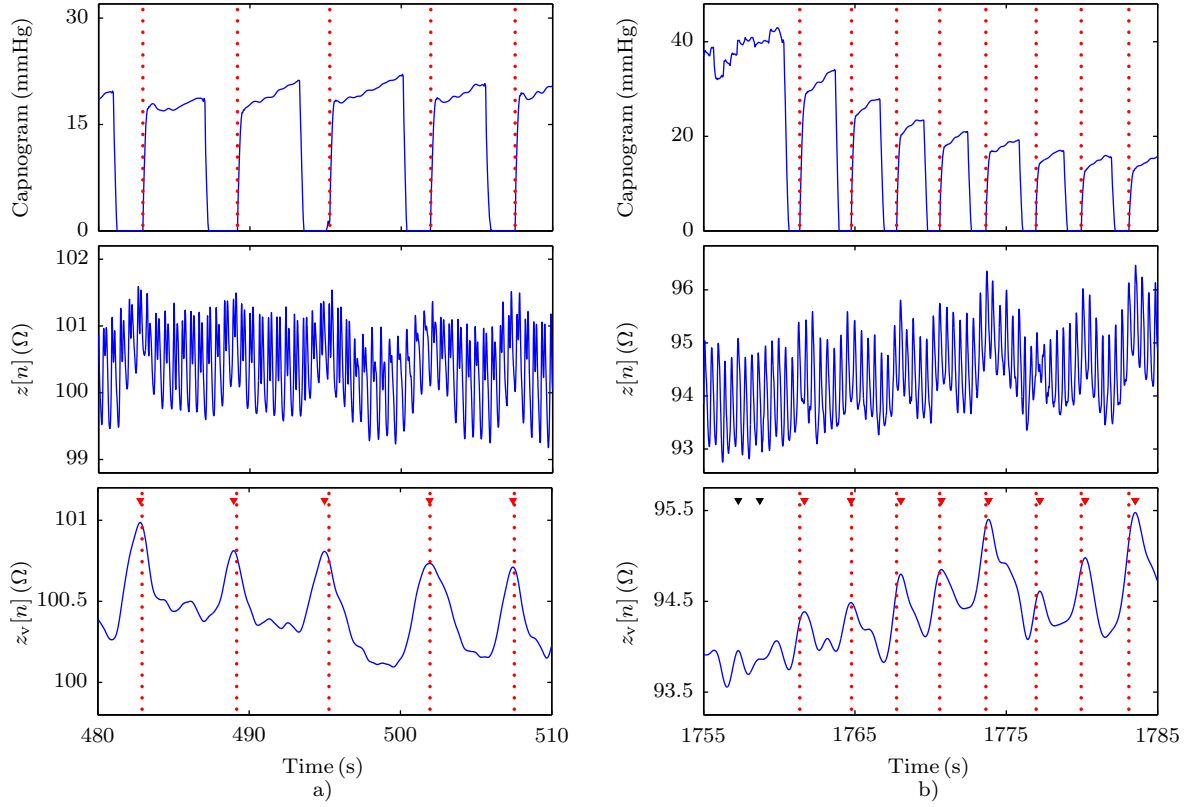


Figure 8: (Supp.) Two different time intervals corresponding to the same episode are shown in panels a and b. The capnography, $z[n]$, and $z_v[n]$ signals can be observed in both panels from top to bottom respectively. Red dotted lines depict the instants of the ventilations marked in the gold standard. Red triangles represent the instants of the correctly detected ventilations (true positives), whereas black triangles represent the instants of the incorrectly detected ventilations (false positives). In $z_v[n]$ of panel a, there are no detection errors. However, there are two detection errors in panel b due to fluctuations in the TI not corresponding to ventilations as can be seen in the capnogram.

389 **Table Legends**

390	Table 1	Mean (SD) values computed from the gold standard and from the TI for
391		each CPR quality metric. The mean (SD) error is also reported. The
392		p – values obtained from the t-test and Mann-Whitney U test (*) are
393		expressed for each metric.

Table 1: Mean (SD) values computed from the gold standard and from the TI for each CPR quality metric. The mean (SD) error is also reported. The p -values obtained from the t-test and Mann-Whitney U test (*) are expressed for each metric.

Metric	Gold Standard	TI	Error	p-value
Mean CC-rate	108.54 (10.54)	109.99 (10.46)	1.75 (4.34)	0.59
FMIR	0.30 (0.26)	0.29 (0.25)	0.03 (0.06)	0.84
CCF	0.81 (0.13)	0.80 (0.13)	0.02 (0.02)	0.71
Mean ventilation rate	9.54 (2.94)	10.87 (2.68)	1.54 (1.44)	0.07
FMH	0.11 (0.18)	0.10 (0.17)	0.01 (0.03)	0.66*
Instantaneous CC-rate	107.42 (16.33)	107.37 (16.26)	1.16 (1.38)	0.99*
Instantaneous ventilation rate	10.23 (4.29)	13.09 (4.52)	3.30 (2.88)	< 0.001